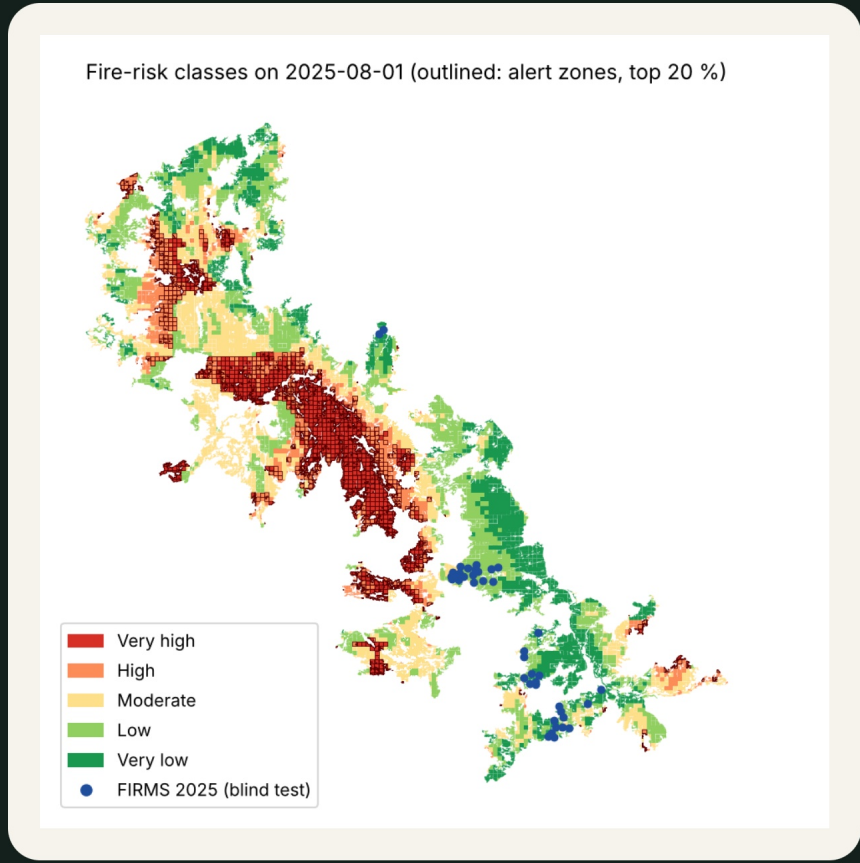
Wiqaya
Satellite Intelligence

ARAB YOUTH SPACE HACKATHON 2026 · FINAL PITCH

See the fire risk from space, before the first spark

Wildfire risk maps and alerts from satellite data, for the Bouhachem, Derdara and Jbel Sougna forests of northern Morocco

Team Wiqaya, Morocco · Climate & Disaster Resilience



Team Overview

Five members across geomatics, AI and web

Member and role	Responsibilities	Skills	Experience
Wiame KANE Team lead · AI model, geomatics	Leads the team Designs the fire-risk model and the spatial data pipeline (grid, features, validation)	Python, GeoPandas, Rasterio, scikit-learn, remote sensing	Geomatics & surveying engineering student, IAV Hassan II
Sanae BELFROUH AI model, API	Model training and its connection to the platform's API	Python, scikit-learn, FastAPI	PhD student in artificial intelligence and neuroscience
Yassine AZMOU AI model, geomatics	Spatial inputs and model experiments	Python, GIS, Earth-observation data	Student in environment and integrated coastal management
Mohammed Yassine SOF Web development	Trilingual web interface and interactive risk map	HTML, CSS, JavaScript, Leaflet	Business intelligence and big data student
Hajar ELOUATIQ Web development	Fire-history calendar, alert sign-up, mobile layout	HTML, CSS, JavaScript	Cybersecurity student

Problem Relevance & Potential Impact

Every summer the Rif burns, and the warning comes too late

≈ 7,500 ha

burned in Bouhachem, July–Aug. 2022 [1]
About 1,100 ha in Sougna, near Derdara, 2021 [2]

445

NASA FIRMS fire detections inside the three forests, 2005–2026 [7, 8, 25]

2

fires on 12–13 Aug. 2025, Derdara and Benqarich [3]
Chergui winds raise fire likelihood [4]

⚠ **The gap:** danger bulletins cover a whole province [5].
Lookouts, thermal drones and satellites only spot a fire once it has started [7, 8, 26, 27].
So the Gendarmerie and Civil Protection are put on alert to fight fires [27].
No one knows which 250 m area needs attention today.

[1] Mongabay, Aug. 2022, news.mongabay.com
[2] Médias24, 19 Aug. 2021, medias24.com
[3] TelQuel, 13 Aug. 2025, telquel.ma
[4] Moumane A. et al., Front. For. Glob. Change 8 (2025)
[5] Hespress, 26 Jun. 2023, fr.hespress.com
[7] Schroeder W. et al., Remote Sens. Environ. 143, 85 (2014)
[8] Giglio L. et al., Remote Sens. Environ. 178, 31 (2016)
[25] Team Wiqaya, github.com/SanaeBelfrouh/wiqayagit (2026)
[26] TelQuel, 30 Jul. 2026, telquel.ma
[27] Hespress, 3 Sep. 2026, en.hespress.com

PROBLEM RELEVANCE & POTENTIAL IMPACT · USERS

Two users: the State and the public

The State

The National Agency for Water and Forests (ANEF), Civil Protection, the Royal Gendarmerie and local authorities.

The public

Villagers living in and around the forest.
People travelling on the roads and tracks between the forests.
Forest visitors.

What Wiqaya gives both

The same 250 m risk map, daily danger level, fire history and alerts.
The only difference: the public signs up first.

Impact at 3 levels: forests, Morocco, Arab region

1 · OUR FORESTS

25,900 ha

Protected today: 6,071 patches of 250 m, each with its own risk [25]

2 · MOROCCO AND THE UN

3 SDGs

UN SDG 13, 15, 11: climate, forests, people [24]

Morocco: supports Forêts du Maroc 2020–2030 [23]

3 · THE ARAB REGION

Reusable

Same data: global, they cover every Arab country [7, 8, 10, 14]

Same method: for any other forest [25]

[7] Schroeder W. et al., Remote Sens. Environ. 143, 85 (2014)
[8] Giglio L. et al., Remote Sens. Environ. 178, 31 (2016)
[10] Hersbach H. et al., Q. J. R. Meteorol. Soc. 146, 1999 (2020)
[14] Haklay M., Weber P., IEEE Pervasive Comput. 7(4), 12 (2008)
[23] Hespress, 23 Jan. 2025, fr.hespress.com
[24] United Nations, Resolution A/RES/70/1 (2015)
[25] Team Wiqaya, github.com/SanaeBelfrouh/wiqayagit (2026)

Scientific & Technical Validity

1 · Data: twenty years of free satellite data, on one grid

Layer	Source	Resolution	Period
Active fires	NASA FIRMS (MODIS, VIIRS) [7, 8]	375 m – 1 km	2005–2026
Terrain	FABDEM V1-2 [9]	30 m	2023 release
Population	WorldPop [6]	100 m	2017
Weather	ERA5 + forecast (Open-Meteo) [10, 30]	Hourly	2005–today
Optical	Sentinel-2 L2A, Landsat C2 L2 [11–13]	10–30 m	2005–2025
Land use	Land-cover map, OpenStreetMap [14, 25]	10 m, vector	2021–2022

All layers in Merchich / Nord Maroc (EPSG:26191)

Open data only: no API key, no imagery cost

[6] Tatem A. J., Sci. Data 4, 170004 (2017)
[7] Schroeder W. et al., Remote Sens. Environ. 143, 85 (2014)
[8] Giglio L. et al., Remote Sens. Environ. 178, 31 (2016)
[9] Hawker L. et al., Environ. Res. Lett. 17, 024016 (2022)
[10] Hersbach H. et al., Q. J. R. Meteorol. Soc. 146, 1999 (2020)
[11] Drusch M. et al., Remote Sens. Environ. 120, 25 (2012)
[12] Wulder M. A. et al., Remote Sens. Environ. 225, 127 (2019)
[13] Key C. H., Benson N. C., USDA RMRS-GTR-164-CD (2006)
[14] Haklay M., Weber P., IEEE Pervasive Comput. 7(4), 12 (2008)
[25] Team Wiqaya, github.com/SanaeBelfrouh/wiqayagit (2026)
[30] Zippenfenig P., Open-Meteo.com Weather API, Zenodo (2024)

2 · Inputs: 16 to know “where”, 4 to know “when”

Each data source gives the model a few inputs, computed for every patch [25].

DEM [9]
Elevation, slope, aspect.

DEM + wind [15]
Topoclimate, index and class.

Land cover 2021 [25]
Land-cover class, flammability.

Sentinel-2 / Landsat [11–13]
NBR of the last image and its trend.

OpenStreetMap [14]
Human pressure: distance to roads, tracks, villages and farmland.

WorldPop [6]
Population per patch.

NASA FIRMS [7, 8]
Fire density and its class, fire power, years since the last fire, burned less than 5 years ago.

Open-Meteo · “when” [10, 19, 30]
Max temperature, min humidity, max wind, max VPD.

[6] Tatem A. J., Sci. Data 4, 170004 (2017)
[7] Schroeder W. et al., Remote Sens. Environ. 143, 85 (2014)
[8] Giglio L. et al., Remote Sens. Environ. 178, 31 (2016)
[9] Hawker L. et al., Environ. Res. Lett. 17, 024016 (2022)
[10] Hersbach H. et al., Q. J. R. Meteorol. Soc. 146, 1999 (2020)
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[13] Key C. H., Benson N. C., USDA RMRS-GTR-164-CD (2006)
[14] Haklay M., Weber P., IEEE Pervasive Comput. 7(4), 12 (2008)
[15] Assali F. et al., Rev. Écol. (Terre Vie) 71(2) (2016)
[19] Allen R. G. et al., FAO Irrigation and Drainage Paper 56 (1998)
[25] Team Wiqaya, github.com/SanaeBelfrouh/wiqayagit (2026)
[30] Zippenfenig P., Open-Meteo.com Weather API, Zenodo (2024)

3 · Method: the model answers “where?” and “when?”

WHERE? Which parts of the forest are fragile

1 · Divide
We divide the forest into small patches of 250 m.

→

2 · Describe
Each patch receives its 16 inputs.

→

3 · Learn
The model learns where fires happened in the past [4, 17].

→

4 · Map
Each patch gets a risk level. The riskiest ones become alert zones.

WHEN? Is today’s weather dangerous

1 · Measure
Every day, we take the 4 weather inputs [30].

→

2 · Compute
The hotter, drier and windier it is, the higher the danger [4].

→

3 · Compare
We compare today with every summer since 2017.

→

4 · Alert
If the danger is high, the alert zones are flagged.

[4] Moumane A. et al., Front. For. Glob. Change 8 (2025)
[17] Pedregosa F. et al., J. Mach. Learn. Res. 12, 2825 (2011)
[30] Zippenfenig P., Open-Meteo.com Weather API, Zenodo (2024)

SCIENTIFIC & TECHNICAL VALIDITY · TESTING

A test on a fire the model has never seen

73 %

Training

4 fires: 2009 (2 fires), 2020 and Derdara 2021.
24,275 patches, 538 of them burned. [25]

27 %

Validation

2 fires: 2022 and Derdara 2025.
8,871 patches, 1,148 of them burned.
Used to choose the model.

1 fire

Test · never seen

Bouhachem, near Derdara, 21 August 2025.
4,905 patches with a valid image, 50 of them burned [13].

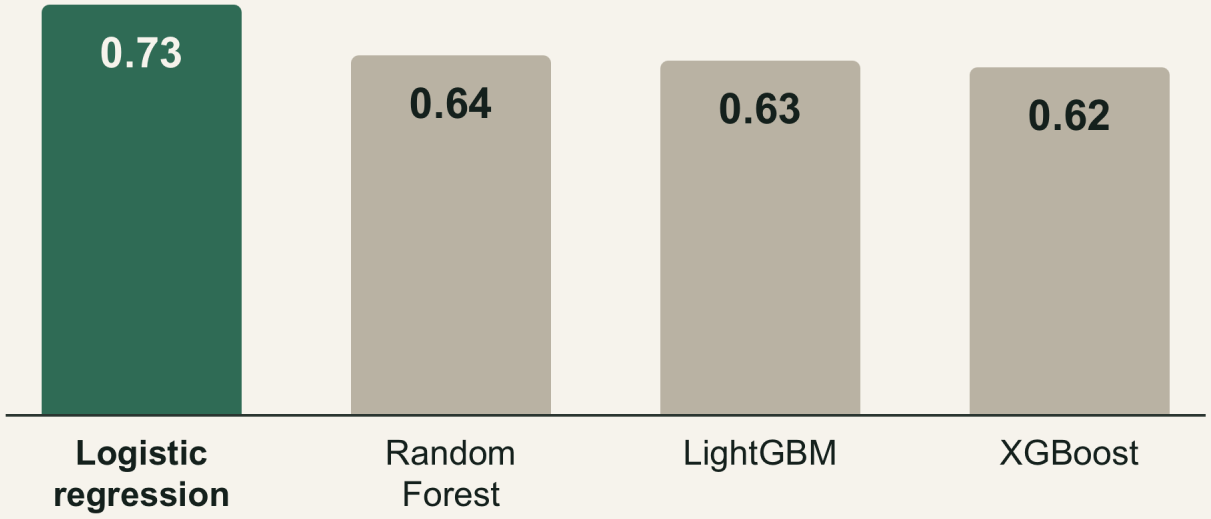
Why these %? We split by fire, not by patch. Two nearby patches of the same fire look alike: mixing them would inflate the score [16].
Of the 6 fires outside the test, 4 are used to learn and 2 to choose the model: 73 % and 27 % of the patches.
This split is repeated 12 times: each fire is the test fire twice.

[13] Key C. H., Benson N. C., USDA RMRS-GTR-164-CD (2006)
[16] Roberts D. R. et al., Ecography 40, 913 (2017)
[25] Team Wiqaya, github.com/SanaeBelfrouh/wiqayagit (2026)

SCIENTIFIC & TECHNICAL VALIDITY · MODEL CHOICE

Four models compared: logistic regression wins

ROC-AUC score of each model on the 2022 validation fire (higher = better). [17, 18, 25, 28, 29]



10 / 12

splits where it is chosen

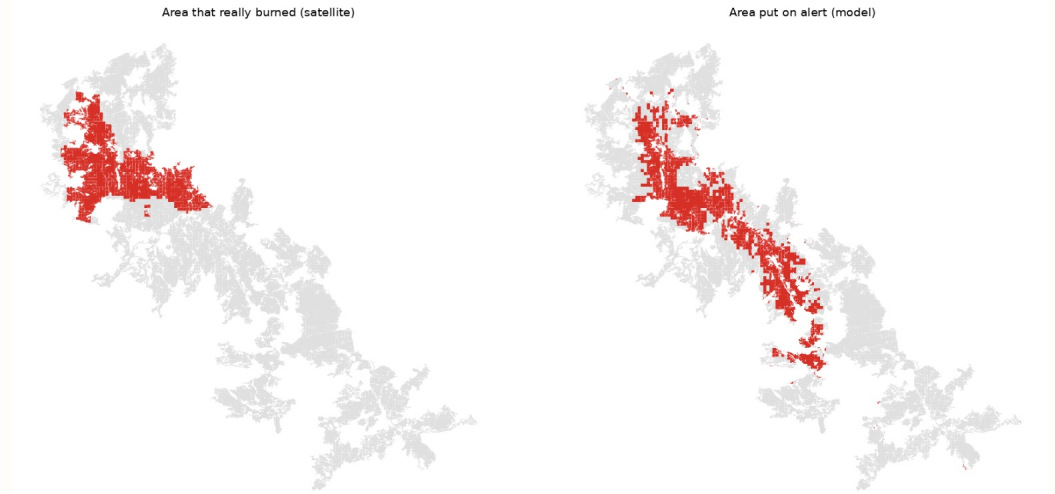
It stays the best when the test fire changes

Why it wins: with only 8 fires, a simple model learns better than a complex one.

[17] Pedregosa F. et al., J. Mach. Learn. Res. 12, 2825 (2011)
[18] Breiman L., Mach. Learn. 45, 5 (2001)
[25] Team Wiqaya, github.com/SanaeBelfrouh/wiqayagit (2026)
[28] Chen T., Guestrin C., Proc. ACM SIGKDD, 785 (2016)
[29] Ke G. et al., Adv. Neural Inf. Process. Syst. 30 (2017)

Tested on a real fire, then used today

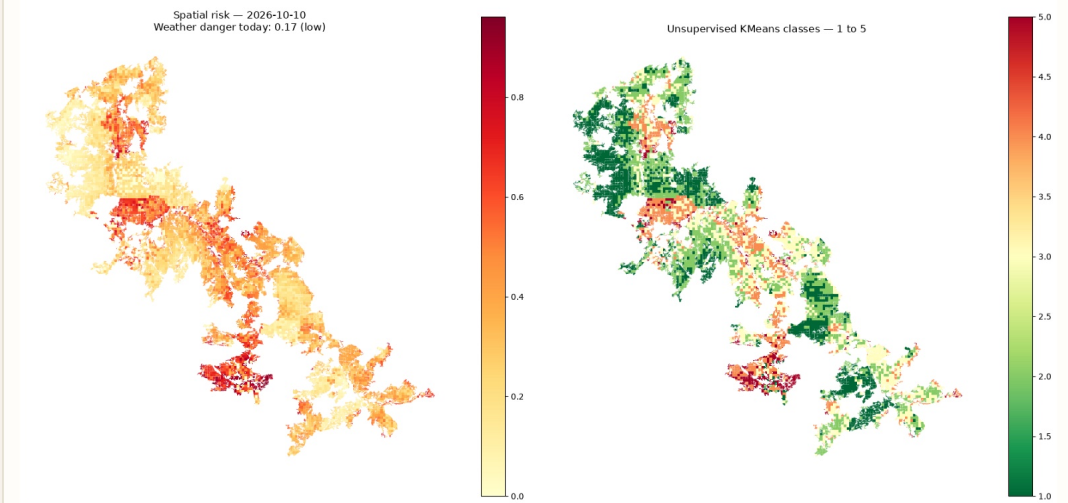
Test · July 2022 fire



0.73 **the best score of the 4 models tested [25]**

Left: what burned [13]. **Right:** the model's alert, which covered **41 %** of the burned area.

Today · 10 October 2026



Low **weather danger today**

In red: the areas to watch first [25, 30].
5 levels: KMeans groups close probabilities, with no thresholds chosen by hand [17].

[13] Key C. H., Benson N. C., USDA RMRS-GTR-164-CD (2006)
[17] Pedregosa F. et al., J. Mach. Learn. Res. 12, 2825 (2011)
[25] Team Wiqaya, github.com/SanaeBelfrouh/wiqayagit (2026)
[30] Zippenfenig P., Open-Meteo.com Weather API, Zenodo (2024)

Confusion matrix

Derdara 2021 fire (test, never seen): alert on the riskiest 20 % of patches [25]

	Model alert	No alert
Burned	209 found	52 missed
Did not burn	1,006 false alarms	4,804 calm, correctly classed

Fire is a rare event

Only **4 %** of patches burned (Derdara 2021).
Never predicting a fire would score **96 %**.

Our 3 scores

80 % — **Recall**: burned patches found

4 × — **Precision**: 17 % of alerts burned, vs 4 % of all patches

0.89 — **ROC-AUC**: out of 1, on a fire never seen

Explanation

1,006 false alarms: areas ready to burn, but no fire started there.
52 missed: a cause our data cannot see, such as a thrown cigarette.

[25] Team Wiqaya, github.com/SanaeBelfrouh/wiqayagit (2026)

Innovation & Differentiation

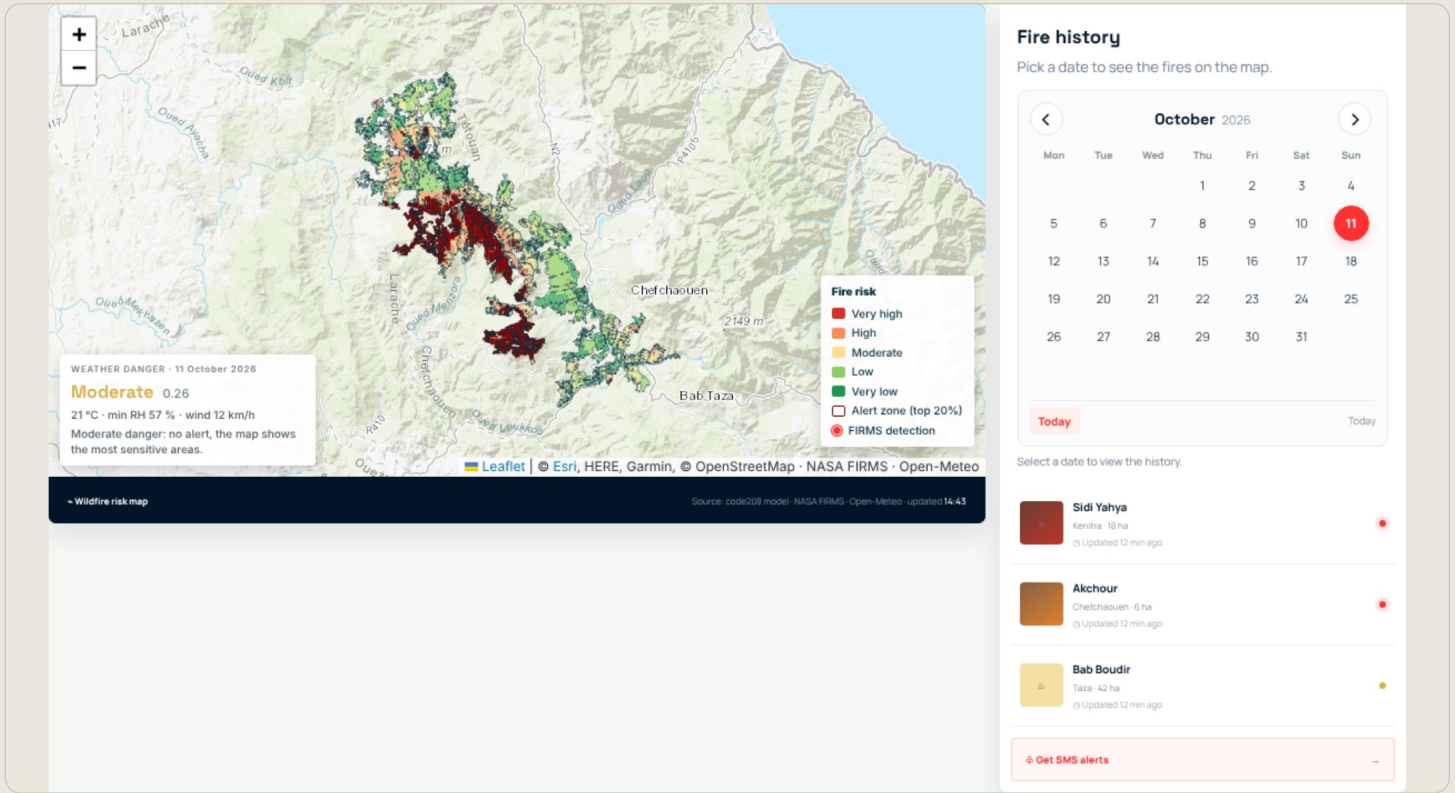
Today, the alert comes late. Wiqaya warns before.

	TODAY	WITH WIQAYA
Scale	**Fixed points** depends on where lookouts and drones are [26, 27]	**250 m** a risk for every patch
Timing	**After** the fire is spotted once it has started [7, 8, 26, 27]	**Before** risky areas are known
For whom	**The State** lookouts and drones [26, 27]	**The State and residents** the same map and the same alerts

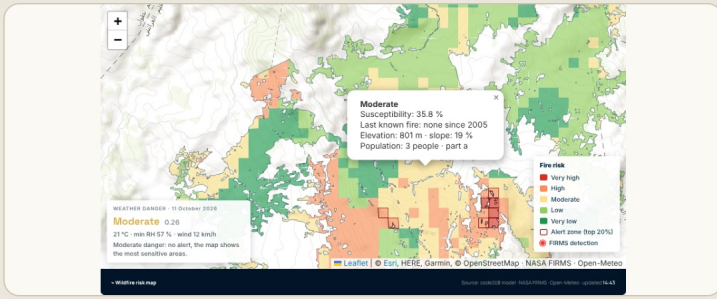
[7] Schroeder W. et al., Remote Sens. Environ. 143, 85 (2014)
[8] Giglio L. et al., Remote Sens. Environ. 178, 31 (2016)
[26] TelQuel, 30 Jul. 2026, telquel.ma
[27] Hespress, 3 Sep. 2026, en.hespress.com

Quality of Implementation

A working prototype, running the same model



Daily risk map with today's weather danger



Each patch explains its risk

MODEL DEPLOYMENT [25]

Model: trained once, then saved on the server. [17]

FastAPI server: takes today's weather (Open-Meteo) and computes the map. [30]

API: sends the 250 m patches as GeoJSON.

Website: Leaflet shows the map, the fire history and the alerts.

[17] Pedregosa F. et al., J. Mach. Learn. Res. 12, 2825 (2011)

[25] Team Wiqaya, github.com/SanaeBelfrouh/wiqayagit (2026)

[30] Zippenfenig P., Open-Meteo.com Weather API, Zenodo (2024)

QUALITY OF IMPLEMENTATION · FOR RESIDENTS

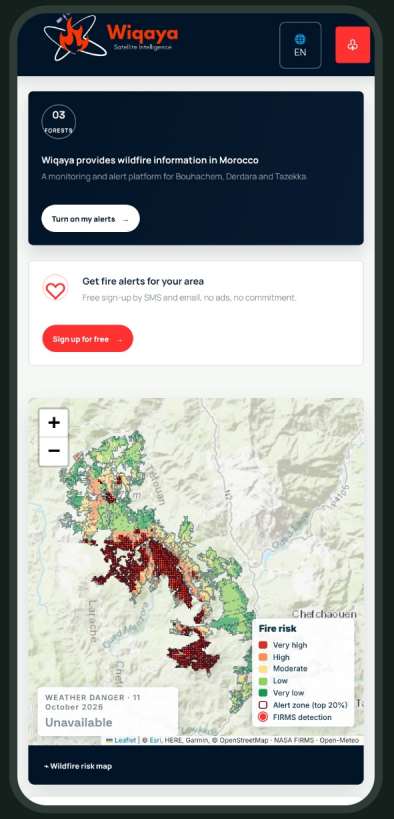
Built for the phone, in three languages

Language: the site opens in the phone's language. Arabic is shown right to left.

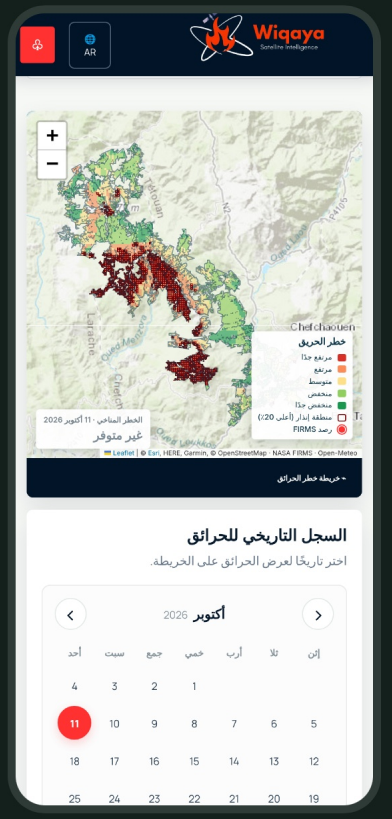
Alerts: sign-up by SMS or email.

History: fires detected by satellite since 2005, day by day. [7, 8]

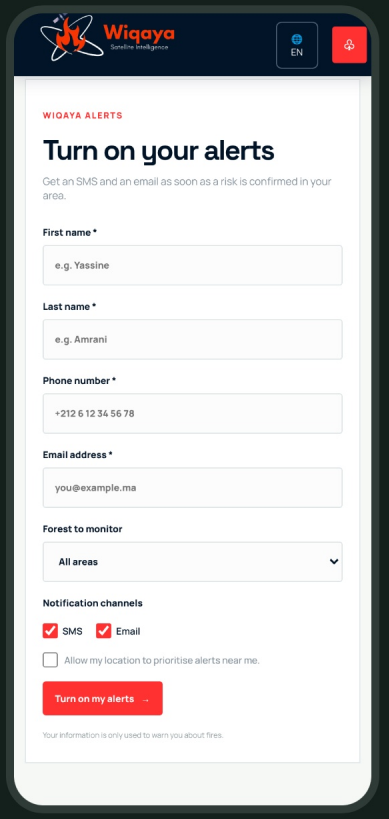
Managers: the map is available as GeoJSON for their GIS software. [25]



Home · English



Map and history · Arabic



Alert sign-up

[7] Schroeder W. et al., Remote Sens. Environ. 143, 85 (2014)

[8] Giglio L. et al., Remote Sens. Environ. 178, 31 (2016)

[25] Team Wiqaya, github.com/SanaeBelfrouh/wiqayagit (2026)

Documentation & Evidence

Everything is open and easy to redo

5 min
to redo everything, on a laptop, no GPU

8.4 MB
of sample data included

MIT
open licence, every dataset credited

- **README:** the project, data, method and limits.
- **Notebook:** the whole computation, start to end.
- **results/:** every map and every score.
- **Inputs PDF:** each model input and its formula.

REDO THE RESULT [25]

```
git clone github.com/
SanaeBelfrouh/wiqayagit
pip install -r requirements.txt
jupyter lab
notebooks/02_main_analysis.ipynb

python server/app.py
# platform on localhost:8000
```

[25] Team Wiqaya, github.com/SanaeBelfrouh/wiqayagit (2026)

Development Potential

Next: from prototype to a fire season

TODAY

A working prototype

Everything shown in this pitch, working locally. [25]

STEP 1

More fires

Add fires from other Rif forests. More examples, a better model. [7, 8]

STEP 2

Hyperspectral

EnMAP or Arab Satellite 813 to measure how dry the plants are. [20–22]

STEP 3

Go online

Host the site, compute the map every day, send the SMS.

STEP 4

Pilot season

One full summer with ANEF and local authorities.

STEP 5

Scale up

Other forests in Morocco, then in the Arab region.

BUSINESS MODEL

Institutions: a subscription per forest (map, API, alerts).

Residents: free alerts, paid for by these subscriptions or CSR partners.

[7] Schroeder W. et al., Remote Sens. Environ. 143, 85 (2014)
[8] Giglio L. et al., Remote Sens. Environ. 178, 31 (2016)
[20] Yebra M. et al., Remote Sens. Environ. 136, 455 (2013)
[21] Guanter L. et al., Remote Sens. 7(7), 8830 (2015)
[22] Khaleej Times, 11 Dec. 2025, khaleejtimes.com
[25] Team Wiqaya, github.com/SanaeBelfrouh/wiqayagit (2026)



Prevention, from space.

Thank you. We are ready for your questions.

</> github.com/SanaeBelfrouh/wiqayagit

Wiame KANE · Sanae BELFROUH · Yassine AZMOU · Mohammed Yassine SOF · Hajar ELOUATIQ

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[3] TelQuel (13 Aug. 2025). Incendies : les forêts de Chefchaouen et Tétouan ravagées par les flammes. telquel.ma/2025/08/13/incendies-les-forets-de-chefchaouen…

[4] Moumane, A. et al. (2025). Advancing wildfire susceptibility mapping through machine learning and SHapley Additive exPlanations-integrated geospatial analysis in Northern Morocco's Mediterranean region. Frontiers in Forests and Global Change, 8. doi.org/10.3389/ffgc.2025.1705341

[5] Hespress (26 Jun. 2023). Incendies de forêts : risque « moyen » à « extrême » dans plusieurs provinces. fr.hespress.com/320358-incendies-de-forets-risque-moyen-a…

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[9] Hawker, L. et al. (2022). A 30 m global map of elevation with forests and buildings removed. Environmental Research Letters, 17, 024016. doi.org/10.1088/1748-9326/ac4d4f

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